

Other Planning Approaches & Sokoban Demo

Video Lesson

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Slide Deck

M10S04 (<https://canvas.tamu.edu/courses/430127/files/92570686?wrap=1>) 

Lesson Transcript

In this lecture, we will look at the remaining issues on planning, alternative approaches, and an interesting game called Sokoban, which is basically a planning game.

Other Formalisms

- Event calculus: Fluents hold at different time points, not situations. Reasoning is done over time.
- Other constructs: generalized events (spatiotemporal), process, intervals, etc.
- Formal theory of belief: propositional attitude, reification, etc.

Aside from situation calculus, which we discussed in detail, and fluent calculus, which we briefly mentioned, there are other formalisms. These include event calculus and other constructs, such as generalized events and formal theory of belief. So, let us look at these one by one.

In event calculus, fluents hold at different time points, not situations, and reasoning is done over time.

Other constructs used for planning include generalized events. These are spatiotemporal events, processes, and intervals.

Formal theory of belief includes propositional attitude, reification, and so on.

Truth Maintenance Systems

New facts inferred from the KB can turn out to be incorrect.

- Let's say P was derived in the KB and later it was found that $\neg P$.
- Adding $\neg P$ to the KB will invalidate the entire KB, so P should be removed ($Retract(KB, P)$).
- Care needs to be taken since other facts in the KB may have been derived from P , etc.
- Truth maintenance systems are designed to handle these complications.

One of the main challenges in planning using knowledge bases is to make sure that new facts that are inferred from the knowledge base are actually true.

When a new inferred fact is false or incorrect, then we can get into trouble. For example, let us say P was derived in the knowledge base, and later on, it was also found that not P is the case. If you have this, and if you put these two facts in the knowledge base, then they will contradict each other, and we are in trouble.

The reason why this is a serious issue is that adding not P to the knowledge base will invalidate the entire knowledge base. So, we must remove this P .

To achieve this, we can invoke the retract function on the knowledge base and tell it to remove this P from the knowledge base. We have to be careful, though, because other facts in the knowledge base may have been derived from that P . So, we have to check. Where are the other places that may have gone wrong?

In general, we have something called the Truth Maintenance System, which is designed to handle these complications.

Planning Approaches

- State-space search: forward or backward.
- Heuristic search: subgoal independence assumption.
- Partial-order planning: utilize problem decomposition. Can place two actions into a plan without specifying the order. Several different total order plans can be constructed from partial order plans.

[image 22 here]

There are several other approaches to planning, including "state-space search," "heuristic search," and "partial-order planning."

In state-space search, you can either search forward or backward. And heuristic search is based on the subgoal independence assumption.

Partial-order planning utilizes problem decomposition. For example, when creating a sequence of actions, you can place two actions into the plan without specifying their relative order.



From this partial order plan, you can construct a total order plan and test which one actually works out.

That concludes this module. Before we end, let us look at an interesting game called Sokoban,

which is basically a planning game.

Here is the Sokoban game. Here is the agent, and you have to push these boxes around and ultimately push all of these boxes into these target locations.

The main difficulty is you can only push. That is, you cannot pull these boxes.

For example, if you push this box once or twice, then you will not be able to move any of these two boxes any further.

Let us see how this player solves this problem.

So again, as you can see, the player is only pushing it, and you must plan carefully.

Yeah, near the end.

And we are done.

Let us look at a much more difficult level.

So here, the task is partially solved.

But as you can see, there are many places where things can go wrong. For example, here, sort of a bottleneck. And I believe some of these boxes were initially here. So again, you have this kind of narrow alleyway, which is very difficult to deal with.

With that, let us conclude this module, and in the next module, we will look at uncertainty and probabilistic reasoning.